Project Asteria – Session Notes

This session covered the Memory System, Experience Learning Engine (ELE), Goal Management System (GMS), and Behaviour Planner (BP).

Memory System

Immediate Memory: last few seconds of sensor/state data.
Short-Term Memory: recent events and actions.
Long-Term Memory: persistent knowledge about the environment.
Memories store confidence values that increase with repeated observations and decay over time if not reinforced.

Experience Learning Engine

Stores Situation → Action → Result → Reward. Positive outcomes increase preference for actions; negative outcomes reduce it. Maintains success/failure statistics and gradually generalizes experience.

Goal Management System

Maintains robot goals with dynamic priorities. Supports pausing/resuming goals using a goal stack, emergency overrides, completion criteria, and failure recovery through replanning.

Behaviour Planner

Converts goals into executable behaviour sequences using reusable behaviours such as Move Forward, Turn, Avoid Obstacle, Search, Dock, etc. Behaviour trees allow flexible execution and interruption handling.

Internal Simulator

Before risky actions, estimate likely outcomes and choose the safest/highest-reward option.

Current High-Level Architecture

Sensors → Sensor Fusion → World Model → Memory System + Experience Learning Engine → Goal Management System → Behaviour Planner → Motor Controller.

Upcoming Modules

Communication Bus, Task Scheduler, Configuration System, Logging & Debugging. After these, the architecture will be frozen and implementation in Python will begin.